# UNIT III
# IoT Data Link Layer and Network Layer Protocols

**Syllabus:** PHY/MAC Layer (3GPP MTC, IEEE 802.11, IEEE 802.15), Wireless HART, Z Wave, Bluetooth Low Energy, Zigbee Smart Energy, DASH7 - Network Layer-IPv4, IPv6, 6LoWPAN, 6TiSCH, ND, DHCP, ICMP, RPL, CORPL, CARP PHY/MAC Layer:

## 3.1. Introduction to PHY/MAC Layer:

In IoT (Internet of Things) architecture, the PHY (Physical) and MAC (Media Access Control) layers refer to the lower layers of the communication stack responsible for establishing and managing the physical connectivity and data transmission between IoT devices.

**PHY Layer:** The PHY layer is the lowest layer in the IoT communication stack and deals with the physical transmission of data over the communication medium. It defines the hardware and electrical specifications required for transmitting and receiving signals. The PHY layer is responsible for aspects such as modulation, coding, frequency bands, transmission power, and signal propagation. It ensures reliable and efficient transmission of data between IoT devices.

Depending on the specific IoT deployment, various wireless technologies can be used at the PHY layer, including Wi-Fi, Bluetooth, Zigbee, Z-Wave, LoRa, Sigfox, Cellular (2G/3G/4G/5G), and others. Each technology has its own PHY layer specifications, such as frequency ranges, transmission power levels, and modulation techniques.

**MAC Layer:** The MAC layer resides above the PHY layer and is responsible for managing access to the shared communication medium and handling data packet transmission between IoT devices. It defines protocols and rules for devices to access the medium, avoiding collisions and ensuring fair and efficient utilization of the available bandwidth.

The MAC layer controls the timing, synchronization, and channel access methods for IoT devices. It can use various techniques such as contention-based access (e.g., CSMA/CA - Carrier Sense Multiple Access with Collision Avoidance), time-based access (e.g., TDMA - Time Division Multiple Access), or scheduled access (e.g., FDMA - Frequency Division Multiple Access) depending on the specific technology being used.

Additionally, the MAC layer may handle functions such as packet fragmentation, error detection, acknowledgments, and retransmissions to ensure reliable data transmission in the presence of interference or other communication challenges.

Both the PHY and MAC layers play a crucial role in establishing robust and efficient communication between IoT devices. They define the physical and access control mechanisms necessary for transmitting data over various wireless technologies, ensuring reliable and optimized connectivity in IoT deployments.

## 3.2. 3GPP MTC:

3GPP MTC (Machine Type Communication) in IoT (Internet of Things) refers to the cellular communication technologies and standards developed by the 3rd Generation Partnership Project (3GPP) specifically for IoT devices. It encompasses the specifications and enhancements made to cellular networks to support the unique requirements of IoT applications.

Here are some key aspects of 3GPP MTC in IoT:

**Low-Power and Low-Complexity Devices:** 3GPP MTC focuses on enabling communication for low-power and low-complexity IoT devices. These devices typically have limited processing capabilities, memory, and power supply, requiring communication technologies that are optimized for efficiency and resource conservation.

**Coverage Enhancement:** 3GPP MTC technologies, such as LTE-M (LTE for MTC), eMTC (enhanced Machine Type Communication), and NB-IoT (Narrowband IoT), offer coverage enhancements to ensure reliable connectivity in challenging environments. These technologies provide extended coverage range, improved signal penetration through buildings and underground areas, and better performance in remote or rural areas.

**Power Efficiency:** Power consumption is a critical concern for IoT devices, many of which operate on battery power. 3GPP MTC introduces power-saving features such as extended discontinuous reception (eDRX) and power saving mode (PSM) to minimize energy consumption. These mechanisms allow IoT devices to enter sleep modes for extended periods while still maintaining network connectivity.

**Optimized Data Rates:** 3GPP MTC technologies provide optimized data rates suitable for IoT applications. While they may not offer the high data throughput of traditional cellular networks, they provide sufficient bandwidth for transmitting small amounts of data, periodic sensor readings, and control messages required by IoT devices.

**Quality of Service (QoS):** 3GPP MTC supports differentiated QoS for different types of IoT applications. It provides mechanisms to prioritize critical data traffic and ensure reliable transmission for mission-critical applications. QoS parameters can be adjusted to meet the specific requirements of diverse IoT use cases.

**Security and Authentication:** 3GPP MTC includes robust security mechanisms to protect IoT device communications. It utilizes encryption, authentication, and access control measures to ensure the confidentiality and integrity of data transmitted over the cellular network.

**Integration with IoT Platforms:** 3GPP MTC technologies are designed to seamlessly integrate with IoT platforms and cloud services. They provide standardized protocols and interfaces for data exchange, device management, and integration with higher-level IoT systems.

By leveraging 3GPP MTC, IoT applications can benefit from the wide coverage, reliable connectivity, power efficiency, and security features offered by cellular networks. These technologies enable cellular communication for a wide range of IoT devices and applications, facilitating the scalability and interoperability of IoT deployments.

Machine-type communication (MTC) is a form of data communication which involves one or more entities that do not necessarily need human interaction.

For MTC communication the following communication scenarios can be identified:

1) Communication scenario with MTC devices communicating with MTC server. MTC server is located in the operator domain

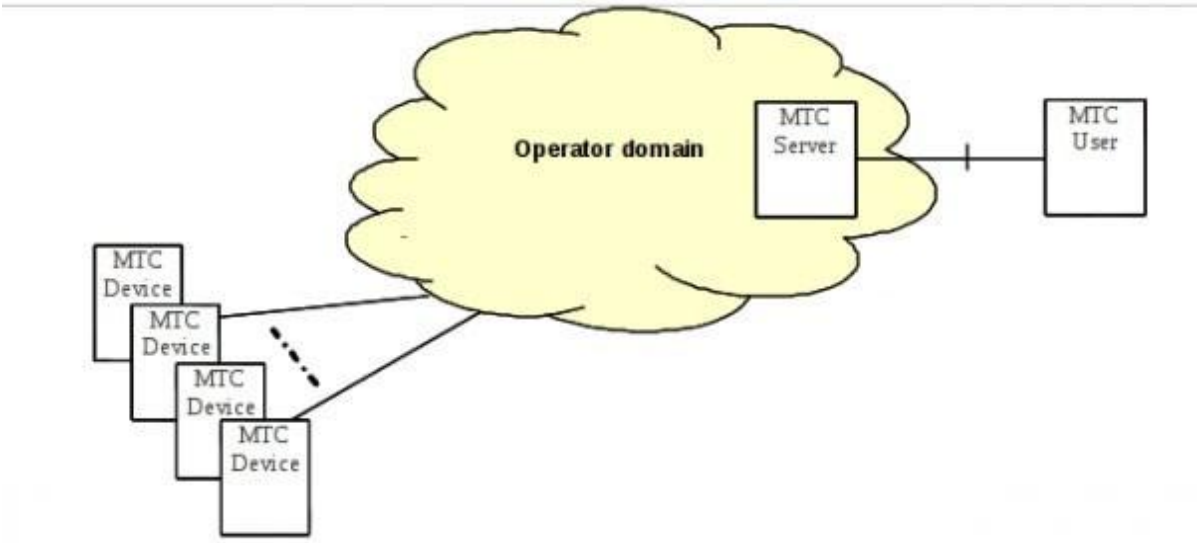



**Fig.3.1. 3GPP MTC Architecture – Indirect Model (MTC Server Controlled by the Network Operator / Inside Operator Domain)**

2) Communication scenario with MTC devices communicating with MTC server. MTC server is located outside the operator domain:

   The network operator provides network connectivity to MTC Server(s). This applies to MTC Servers controlled by the network operator or to MTC Servers not controlled by the network operator.

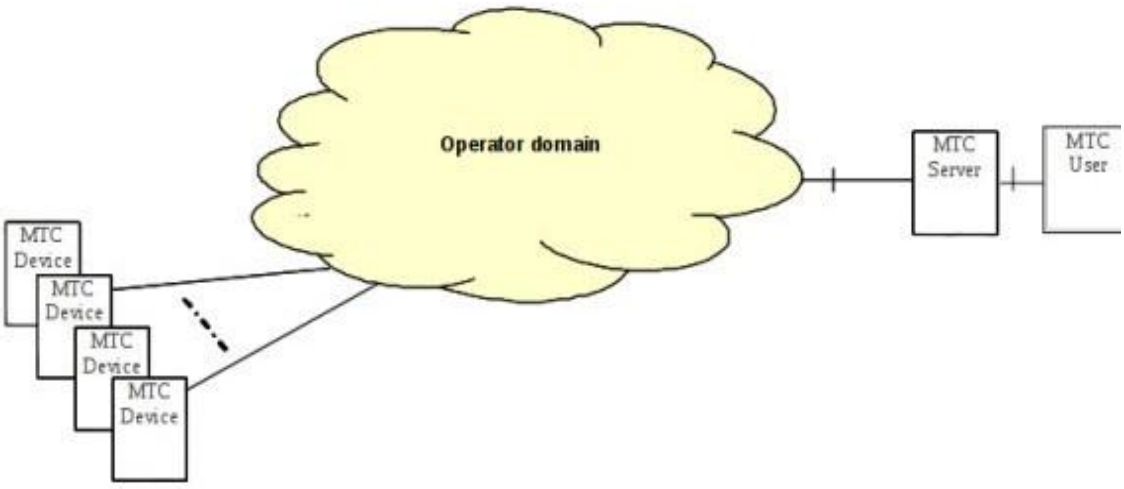



**Fig.3.2: 3GPP Machine Type Communication (MTC) Architecture (Indirect Model / MTC Server controlled by MTC User setup).**

3) MTC devices communicating with each other:
   The communication scenario where the MTC Devices communicate directly without intermediate MTC Server

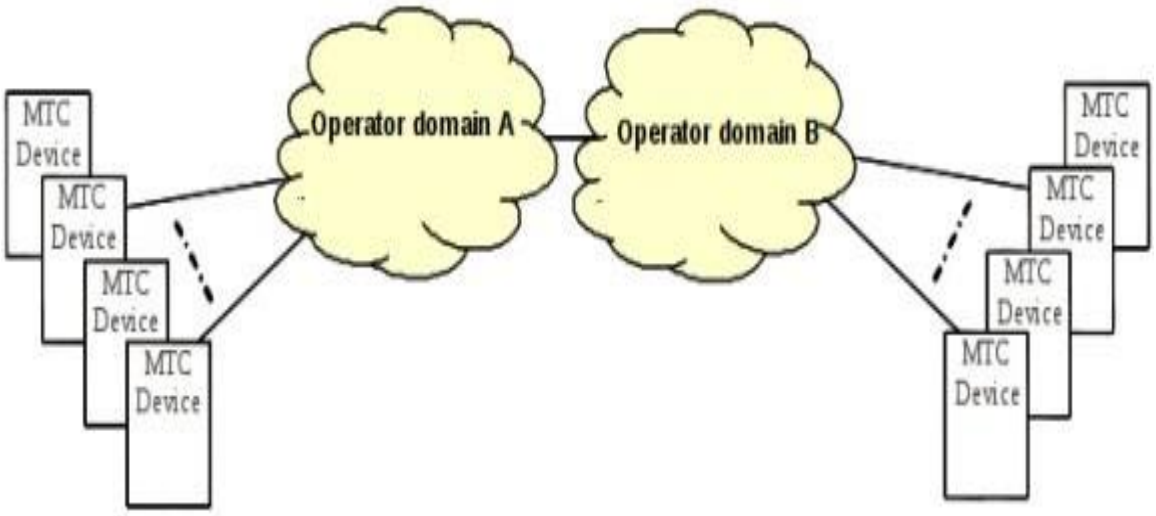



**Fig.3.3: 3GPP MTC Architecture – Direct Model (Device-to-Device / Peer-to-Peer MTC)**.

As technology evolves, there are important changes in capabilities and costs. More computing power, memory and communication capabilities make it possible for machines to take over tasks presently done by, but not well suited to human beings. Lower costs make it practical for machines to take over tasks not well suited to expensive human beings. Increasing capabilities and lower costs together open new opportunities for revenue generating services not previously economical to do.

The increasing capability of machines makes it possible to avoid dull and repetitious work having to be done by people, freeing them to utilize their capabilities and intelligence in better suited and much more fruitful activities.

**MTC Device:** A MTC Device is a UE equipped for Machine Type Communication, which communicates through a PLMN with MTC Server(s) and/or other MTC Device(s).

**MTC Feature:** MTC Features are network functions to optimise the network for use by M2M applications.

**MTC Group:** A MTC Group is a group of MTC Devices that share one or more MTC Features and that belong to the same MTC Subscriber.

**MTC Server:** A MTC Server is a server, which communicates to the PLMN itself, and to MTC Devices through the PLMN. The MTC Server can also have an interface which can be accessed by the MTC User. The MTC Server can:

Provides services for other servers (e.g. The MTC Server is a Services Capability Server for an Application Server), and/or

Provides services for applications and can host the application (e.g. The MTC Server is an Application Server).

**MTC User:** A MTC User uses the service provided by the MTC Server.

**MTC Subscriber:** A MTC Subscriber is a legal entity having a contractual relationship with the network operator to provide service to one or more MTC Devices.

## 3.3. IEEE 802.11 STANDARDS ( WLAN) :

IEEE 802.11 standard, popularly known as WiFi, lays down the architecture and specifications of wireless LANs (WLANs). WiFi or WLAN uses high frequency radio waves for connecting the nodes.

### 3.3.1. Classification of IEEE 802.11

There are several standards of IEEE 802.11 WLANs. The prominent among them are 802.11, 802.11a, 802.11b, 802.11g, 802.11n and 802.11p. All the standards use carrier-sense multiple access with collision avoidance (CSMA/CA). Also, they have support for both centralized base station based as well as ad hoc networks.

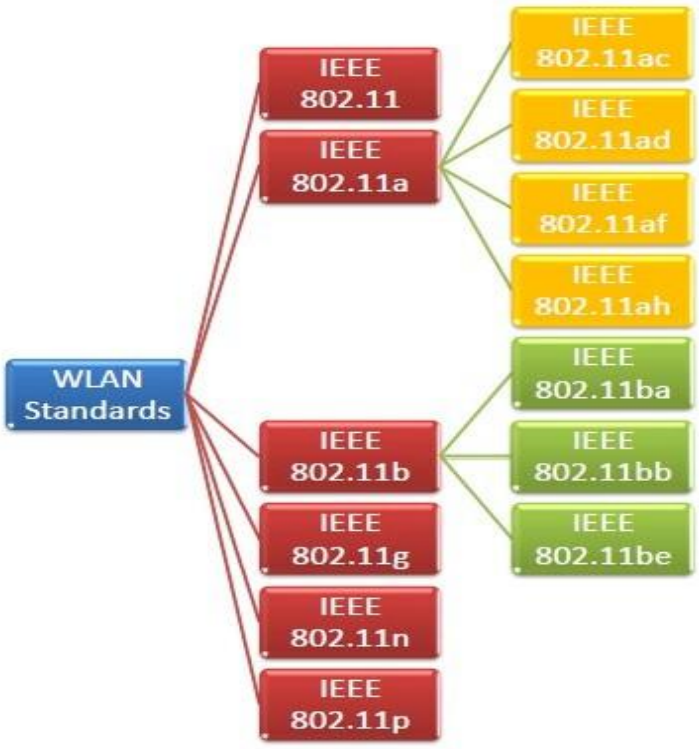



**Fig.3.4: Classification of IEEE 802.11 WLAN Standards**

**IEEE 802.11**

IEEE 802.11 was the original version released in 1997. It provided 1 Mbps or 2 Mbps data rate in the 2.4 GHz band and used either frequency-hopping spread spectrum (FHSS) or direct-sequence spread spectrum (DSSS). It is obsolete now.

**IEEE 802.11a**

802.11a was published in 1999 as a modification to 802.11, with orthogonal frequency division multiplexing (OFDM) based air interface in physical layer instead of FHSS or DSSS of 802.11. It provides a maximum data rate of 54 Mbps operating in the 5 GHz band. Besides it provides error correcting code. As 2.4 GHz band is crowded, relatively sparsely used 5 GHz imparts additional advantage to 802.11a.

Further amendments to 802.11a are 802.11ac, 802.11ad, 802.11af, 802.11ah, 802.11ai, 802.11aj etc.

**IEEE 802.11b**

802.11b is a direct extension of the original 802.11 standard that appeared in early 2000. It uses the same modulation technique as 802.11, i.e. DSSS and operates in the 2.4 GHz band. It has a higher data rate of 11 Mbps as compared to 2 Mbps of 802.11, due to which it was rapidly adopted in wireless LANs. However, since 2.4 GHz band is pretty crowded, 802.11b devices faces interference from other devices.

Further amendments to 802.11b are 802.11ba, 802.11bb, 802.11bc, 802.11bd and 802.11be.

**IEEE 802.11g**

802.11g was indorsed in 2003. It operates in the 2.4 GHz band (as in 802.11b) and provides a average throughput of 22 Mbps. It uses OFDM technique (as in 802.11a). It is fully backward compatible with 802.11b. 802.11g devices also faces interference from other devices operating in 2.4 GHz band.

**IEEE 802.11n**

802.11n was approved and published in 2009 that operates on both the 2.4 GHz and the 5 GHz bands. It has variable data rate ranging from 54 Mbps to 600 Mbps. It provides a marked improvement over previous standards 802.11 by incorporating multiple-input multiple-output antennas (MIMO antennas).

**IEEE 802.11p**

802.11 is an amendment for including wireless access in vehicular environments (WAVE) to support Intelligent Transportation Systems (ITS). They include network communications between vehicles moving at high speed and the environment. They have a data rate of 27 Mbps and operate in 5.9 GHz band.

### 3.3.2. IEEE 802.11 protocol architecture:

Wireless LANs are those Local Area Networks that use high frequency radio waves instead of cables for connecting the devices in LAN. Users connected by WLANs can move around within the area of network coverage. Most WLANs are based upon the standard IEEE 802.11 or WiFi.

The components of IEEE 802.11 architecture are as follows

**1) Stations (STA)** − Stations comprise all devices and equipments that are connected to the wireless LAN. A station can be of two types:

Wireless Access Points (WAP)−WAPs or simply access points (AP) are generally wireless routers that form the base stations or access.

Client− Clients are workstations, computers, laptops, printers, smart phones, etc. Each station has a wireless network interface controller.

**2) Basic Service Set (BSS)** −A basic service set is a group of stations communicating at physical layer level. BSS can be of two categories depending upon mode of operation:

Infrastructure BSS − Here, the devices communicate with other devices through access points.

Independent BSS − Here, the devices communicate in peer-to-peer basis in an ad hoc manner.

**3) Extended Service Set (ESS)** − It is a set of all connected BSS.

**4) Distribution System (DS)** − It connects access points in ESS**.**

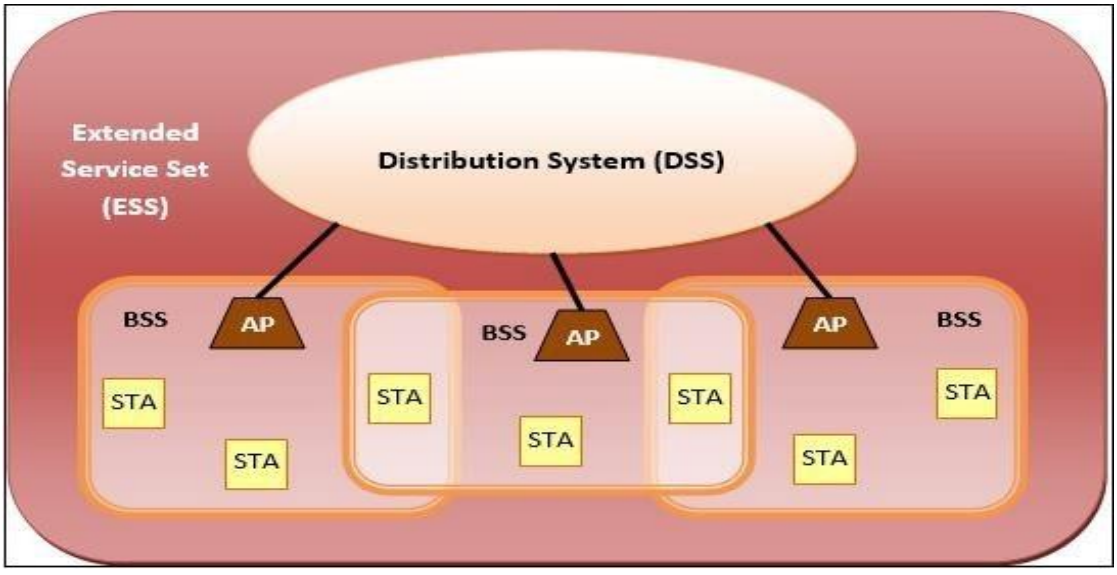



**Fig.3.5: Architecture of IEEE 802.11**

An ESS can also provide gateway access for wireless users into a wired network. Each end station associates itself with one access point. Above Figure shows three BSSs interconnected through three APs to a distribution system. If station A associated with AP-1 wants to send a frame to another station associated with AP- 2, the first sends a frame to its access point (AP-1), which forwards the frame across the distribution system to the access point AP-2. AP-2 finally delivers it to the destination station.
The technique used for this purpose is known as scanning, which involves the following steps:

- A station sends a probe frame.
- All APs within reach reply with a probe response frame.
- The station selects one of the access points, and sends the AP an Association Request frame.
- The AP replies with an Association Response frame.

### 3.3.3. Medium Access Control:

Most wired LANs products use Carrier Sense Multiple Access with Collision Detection (CSMA/CD) as the MAC protocol. Carrier Sense means that the station will listen before it transmits. If there is already someone transmitting, then the station waits and tries again later. If no one is transmitting then the station goes ahead and sends what it has. But when more than one station tries to transmit, the transmissions will collide and the information will be lost.

This is where Collision Detection comes into play. The station will listen to ensure that its transmission made it to the destination without collisions. If a collision occurred then the stations wait and try again later. The time the station waits is determined by the back off algorithm. This technique works great for wired LANs but wireless topologies can create a problem for CSMA/CD. However, the wireless medium presents some unique challenges not present in wired LANs that must be dealt with by the MAC used for IEEE
802.11. Some of the challenges are:

- The wireless LAN is prone to more interference and is less reliable.
- The wireless LAN is susceptible to unwanted interception leading to security problems.
- There are so called hidden station and exposed station problems.

In the discussion of both the problem, we shall assume that all radio transmitters have fixed range. When the receiver is in the range of two active transmitters then the signal will be garbled. It is important to note that not all stations are in range of two transmitters.

### 3.3.4. The Hidden Station Problem

Consider a situation when A is transmitting to B, as depicted in the Fig. If C senses the media, it will not hear anything because it is out of range, and thus will falsely conclude that no transmission is going on and will start transmit to
B. the transmission will interfere at B, wiping out the frame from A.

The problem of a station not been able to detect a potential competitor for the medium because the competitor is too far away is referred as Hidden Station Problem. As in the described scenario C act as a hidden station to A, this is also competing for the medium.

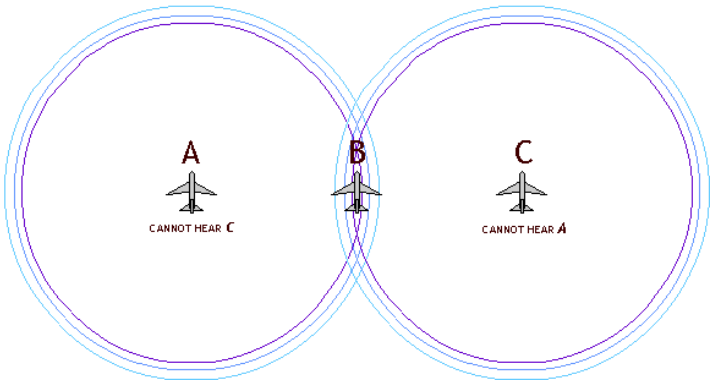



**Fig.3.6: Hidden Station Problem in IEEE 802.11**

### 3.3.5. Exposed Station problem

Now consider a different situation where B is transmitting to A, and C sense the medium and detects the ongoing transmission between B and A. C falsely conclude that it cannot transmit to D, when the fact is that such transmission would cause on problem. A transmission could cause a problem only when the destination is in zone between B and C. This problem is referred as Exposed station Problem. In this scenario as B is exposed to C, that's why C assumes it cannot transmit to D. So this problem is known as Exposed station problem (i.e. problem caused due to exposing of a station). The problem here is that before transmission, a station really wants to know that whether or not there is any activity around the receiver. CSMA merely tells whether or not there is any activity around the station sensing the carrier.

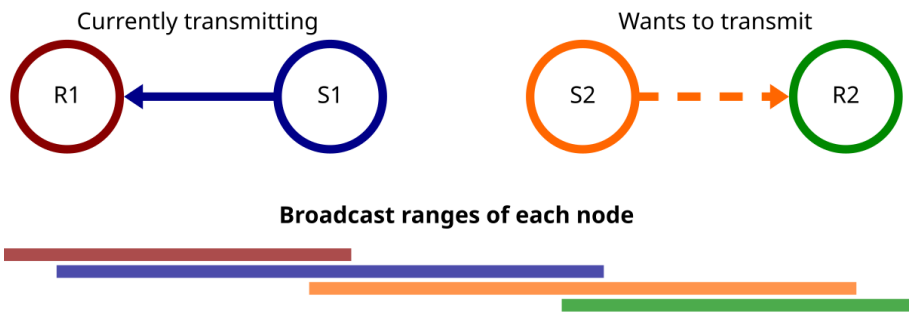



**Fig.3.7. Exposed Node Problem in IEEE 802.11**

### 3.3.6. Security

Wireless LANs are subjected to possible breaches from unwanted monitoring. To overcome this problem, IEEE 802.11 specifies an optional MAC layer security system known as Wired Equivalent Privacy (WEP). The objective is to provide a level of privacy to the wireless LAN similar to that enjoyed by wired Ethernets. It is achieved with the help of a 40-bit shared key authentication service. By default each BSS supports up to four 40-bit keys that are shared by all the clients in the BSS. Keys unique to a pair of communicating clients and direction of transmission may also be used. Advanced Encryption Standard (AES) (802.11i) for authentication and encryption is recommended as a long-term solution.

### 3.3.7. Frame Format of IEEE 802.11

The main fields of a frame of wireless LANs as laid down by IEEE 802.11 are −

Frame Control − It is a 2 bytes starting field composed of 11 subfields. It contains control information of the frame.

Duration − It is a 2-byte field that specifies the time period for which the frame and its acknowledgment occupy the channel.

Address fields − There are three 6-byte address fields containing addresses of source, immediate destination, and final endpoint respectively.

Sequence − It a 2 bytes field that stores the frame numbers.

Data − This is a variable-sized field that carries the data from the upper layers. The maximum size of the data field is 2312 bytes.

Check Sequence − It is a 4-byte field containing error detection information.

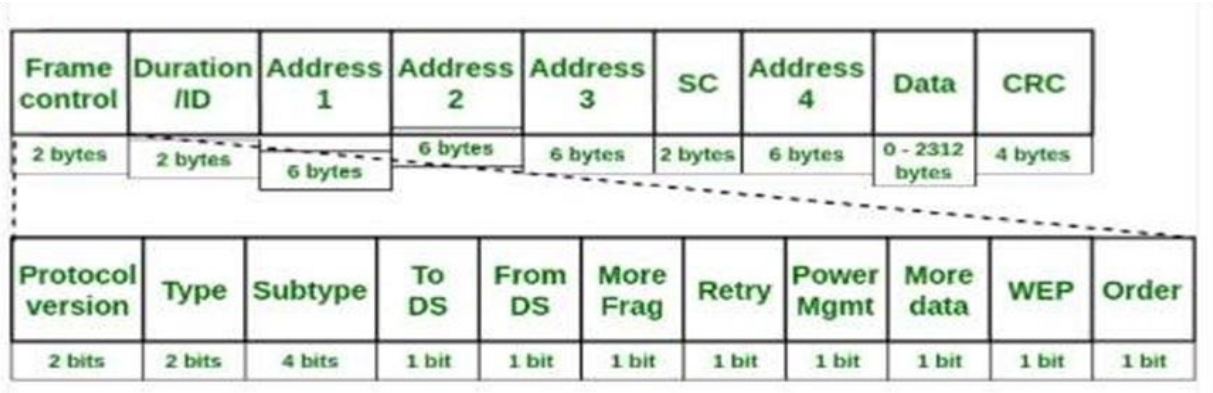



**Fig.3.8. IEEE 802.11 Frame Structure**

**Frame Control(FC) –**

It is 2 bytes long field which defines type of frame and some control information. Various fields presentin FC are:

1. **Version:**

It is a 2 bit long field which indicates the current protocol version which is fixed to be 0 for now.

**2. Type:**

It is a 2 bit long field which determines the function of frame i.e management (00), control (01) ordata (10). The value 11 is reserved.

**3. Subtype:**

It is a 4 bit long field which indicates sub-type of the frame like 0000 for association request, 1000 forbeacon.

**4. To DS:**

It is a 1 bit long field which when set indicates that destination frame is for DS(distribution system).

**5. From DS:**

It is a 1 bit long field which when set indicates frame coming from DS.

**6. More frag (More fragments):**

It is 1 bit long field which when set to 1 means frame is followed by other fragments.

**7. Retry:**

It is 1-bit long field, if the current frame is a retransmission of an earlier frame, this bit is set to 1.

**8. Power Mgmt (Power management):**

It is 1-bit long field that indicates the mode of a station after successful transmission of a frame. Set to 1 the field indicates that the station goes into power-save mode. If the field is set to 0, the station stays active.

**9. More data:**

It is 1-bit long field that is used to indicate receiver that a sender has more data. to send than the current frame. This can be used by an access point to indicate to a station in power-save mode that more packets are buffered or it can be used by a station to indicate to an access point after being polled that more polling is necessary as the station has more data ready to transmit.

**10. WEP:**

It is 1 bit long field which indicates that the standard security mechanism of 802.11 is applied.

**11. Order:**

It is 1 bit long field, if this bit is set to 1 the received frames must be processed in strict order.

## 3.4. IEEE 802.15 Standards ( WPAN/ WSN):

IEEE 802.15 is a standard for wireless personal area networks (WPANs) or Wireless

Specialty Networks(WSNs) developed by the Institute of Electrical and Electronics Engineers (IEEE). It defines the physical layer (PHY) and medium access control (MAC) layer specifications for short-range wireless communication.

### 3.4.1. IEEE 802.15 Family

The IEEE 802.15 architecture consists of multiple task groups, each focusing on specific applications and requirements. Here are some key task groups within the IEEE 802.15 standard:

**IEEE 802.15.1 (Bluetooth):** This task group defines the specifications for Bluetooth wireless technology, which enables short-range communication between devices such as mobile phones, laptops, and peripherals.

**IEEE 802.15.3:** This task group focuses on high-rate wireless personal area networks (HR-WPANs). It defines the PHY and MAC layers for applications that require higher data rates, such as streaming multimedia.

**IEEE 802.15.4:** This task group specifies the PHY and MAC layers for low-rate wireless personal area networks (LR-WPANs). It is commonly used in applications like home automation, industrial control, and wireless sensor networks. The most well-known standard built on IEEE 802.15.4 is Zigbee.

**IEEE 802.15.6:** This task group concentrates on wireless body area networks (WBANs). It addresses the specific requirements of medical, healthcare, and fitness applications by defining the PHY and MAC layers suitable for wearable and implantable devices.

**IEEE 802.15.7:** This task group defines the PHY and MAC layers for visible light communication (VLC). It enables communication using light-emitting diodes (LEDs) and is often used for indoor positioning, smart lighting, and other applications.

### 3.4.2. IEEE 802.15.4

IEEE 802.15.4 is a subgroup of features that refers to physical and medium access control layers that can support ZigBee and 6LoWPAN upper. IEEE 802.15.4 focuses on physical and data link layer specifications while ZigBee Alliance aims to provide the upper characteristics [33]. It is a standard that defines PHY and MAC layers for personal area networks that demand low rate and low cost applications. This also called a LR-WPAN protocol and has some advantages. Among them are a simple and flexible protocol stack, low cost, low energy consumption, short-range operation, reliable data transfer, and ease of operation [34]. These features are more important when operating in the Personal Operating Space (POS) also defined as Personal Area Network (PAN) that involves the human body.

Physical Layer

The 802.15.4 standard supports an extensive number of PHY options that range from

2.4 GHz to sub-GHz frequencies in ISM bands. (ISM bands are discussed earlier in this chapter.) The original IEEE 802.15.4-2003 standard specified only three PHY options based on direct sequence spread spectrum (DSSS) modulation. DSSS is a modulation technique in which a signal is intentionally spread in the frequency domain, resulting in greater bandwidth. The original physical layer transmission options were as follows:

- 2.4 GHz, 16 channels, with a data rate of 250 kbps
- 915 MHz, 10 channels, with a data rate of 40 kbps
- 868 MHz, 1 channel, with a data rate of 20 kbps

You should note that only the 2.4 GHz band operates worldwide. The 915 MHz band operates mainly in North and South America, and the 868 MHz frequencies are used in Europe, the Middle East, and Africa. IEEE 802.15.4-2006, 802.15.4-2011, and

IEEE 802.15.4-2015 introduced additional PHY communication options, including the following:

- OQPSK PHY: This is DSSS PHY, employing offset quadrature phase-shift keying (OQPSK) modulation. OQPSK is a modulation technique that uses four unique bit values that are signaled by phase changes. An offset function that is present during phase shifts allows data to be transmitted more reliably.

- BPSK PHY: This is DSSS PHY, employing binary phase-shift keying (BPSK) modulation. BPSK specifies two unique phase shifts as its data encoding scheme.

- ASK PHY: This is parallel sequence spread spectrum (PSSS) PHY, employing amplitude shift keying (ASK) and BPSK modulation. PSSS is an advanced encoding scheme that offers increased range, throughput, data rates, and signal integrity compared to DSSS. ASK uses amplitude shifts instead of phase shifts to signal different bit values.

These improvements increase the maximum data rate for both 868 MHz and 915 MHz to 100 kbps and 250 kbps, respectively. The 868 MHz support was enhanced to 3 channels, while other IEEE 802.15.4 study groups produced addendums for new frequency bands. For example, the IEEE 802.15.4c study group created the bands 314–316 MHz, 430–434 MHz, and 779–787 MHz for use in China.

Figure shows the frame for the 802.15.4 physical layer. The synchronization header for this frame is composed of the Preamble and the Start of Frame Delimiter fields. The Preamble field is a 32-bit 4-byte (for parallel construction) pattern that identifies the start of the frame and is used to synchronize the data transmission. The Start of

Frame Delimiter field informs the receiver that frame contents start immediately after this byte.

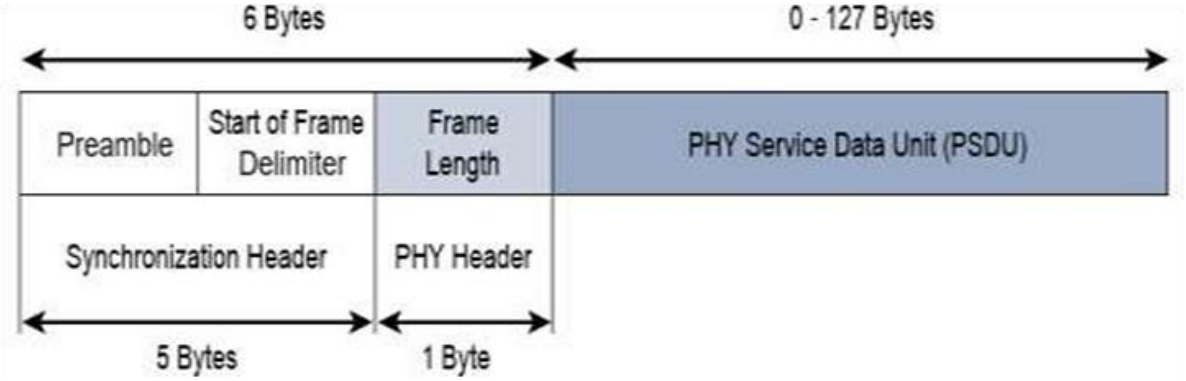



**Fig.3.9: PHY header**

The PHY Header portion of the PHY frame shown in Figure 4-5 is simply a frame length value. It lets the receiver know how much total data to expect in the PHY service data unit (PSDU) portion of the 802.4.15 PHY. The PSDU is the data field or payload.

**MAC Layer**

The IEEE 802.15.4 MAC layer manages access to the PHY channel by defining how devices in the same area will share the frequencies allocated. At this layer, the scheduling and routing of data frames are also coordinated. The 802.15.4 MAC layer performs the following tasks:

- Network beaconing for devices acting as coordinators (New devices use beacons to join an 802.15.4 network)
- PAN association and disassociation by a device
- Device security
- Reliable link communications between two peer MAC entities

The MAC layer achieves these tasks by using various predefined frame types. In fact, four types of MAC frames are specified in 802.15.4:

- Data frame: Handles all transfers of data
- Beacon frame: Used in the transmission of beacons from a PAN coordinator
- Acknowledgement frame: Confirms the successful reception of a frame
- flAC command frame: Responsible for control communication between devices

Each of these four 802.15.4 MAC frame types follows the frame format shown in Figure . In Figure, notice that the MAC frame is carried as the PHY payload. The

802.15.4 MAC frame can be broken down into the MAC Header, MAC Payload, and MAC Footer fields.

The MAC Header field is composed of the Frame Control, Sequence Number and the Addressing fields. The Frame Control field defines attributes such as frame type, address- ing modes, and other control flags. The Sequence Number field indicates the sequence identifier for the frame. The Addressing field specifies the Source and Destination PAN Identifier fields as well as the Source and Destination Address fields.

The MAC Payload field varies by individual frame type. For example, beacon frames have specific fields and payloads related to beacons, while MAC command frames have different fields present. The MAC Footer field is nothing more than a frame check sequence (FCS). An FCS is a calculation based on the data in the frame that is used by the receiving side to confirm the integrity of the data in the frame. IEEE 802.15.4 requires all devices to support a unique 64-bit extended MAC address, based on EUI-64. However, because the maximum payload is 127 bytes, 802.15.4 also defines how a 16-bit "short address" is assigned to devices. This short address is local to the PAN and substantially reduces the frame overhead compared to a 64-bit extended MAC address. However, you should be aware that the use of this short address might be limited to specific upper-layer protocol stacks.

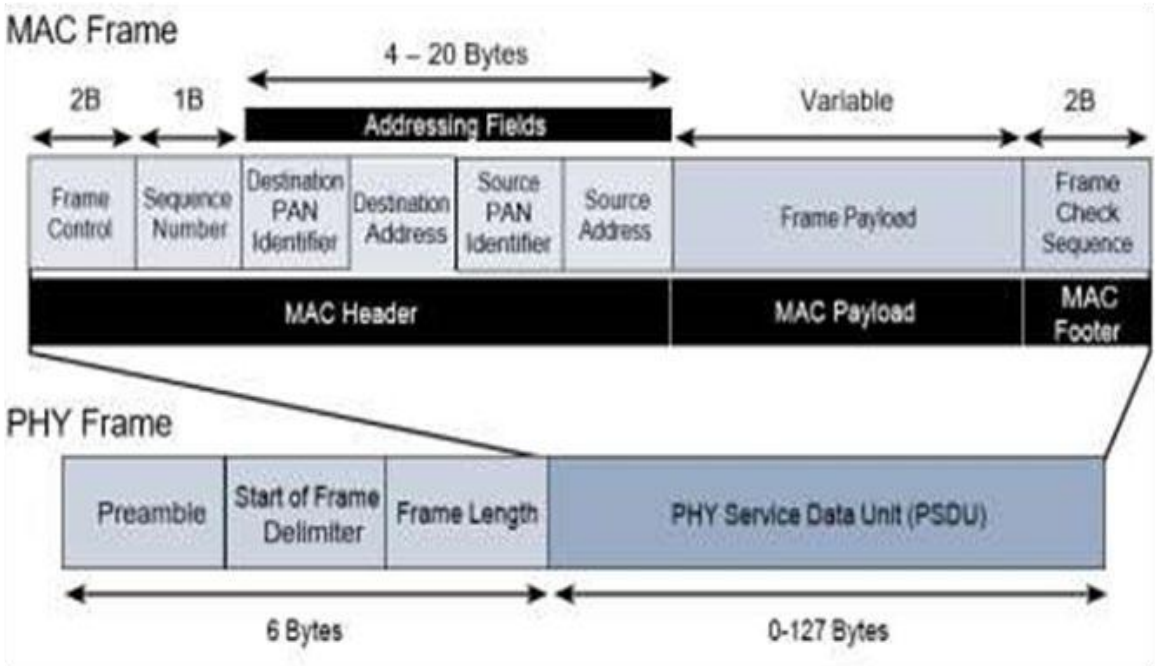



**Fig.3.10: IEEE 802.15.4 MAC Format**

The IEEE 802.15.4 BE and NBE operational modes have being strongly investigated over recent years. Thus, some limitations have been addressed and the most important ones are the unbounded delay, low communication efficiency, low interference robustness, and/or fading and main powered relay nodes Figure compares IEEE 802.15.4 stack with the OSI reference model.

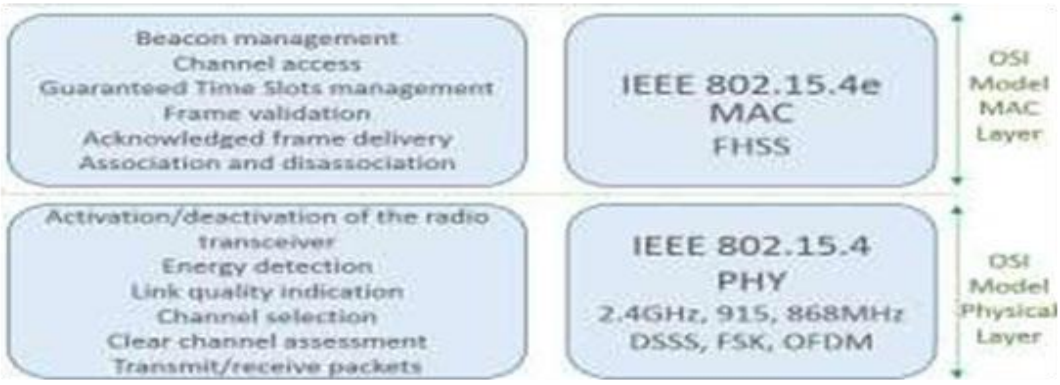



**Fig.3.11: IEEE 802.15.4 compared with OSI reference model.**

**Advantages of IEEE 802.15.4:**

IEEE 802.15.4 has the following advantages:

-cheap cost

-long battery life,

-Quick and simple installation

-extensible protocol stack

**Disadvantages of IEEE 802.15.4:**

IEEE 802.15.4's drawbacks include:

-IEEE 802.15.4 causes interference and multipath fading.

-doesn't employ a frequency-hopping approach.

-unbounded latency

-interference susceptibility

**Applications of IEEE 802.15.4:** IEEE

802.15.4 Applications:

-Wireless sensor networks in the industry

-Building and home automation

-Remote controllers and interacting toys

-Automotive networks